

Technical Test Senior Data Engineer

Hi there,

Thank you for applying to Stockbit/Bibit for the Data Engineer position. Before we schedule a conversation with you, we'd like you to complete this challenge to help us understand your skills and approach better.

For this test, we're providing the "Brazilian E-Commerce Public Dataset by Olist" from Kaggle. You'll find the dataset in CSV format within the provided folder, along with an ERD to help you understand the data model. We've excluded the [olist_geolocation_dataset](#) since it only contains geolocation data and isn't essential for this challenge.

This dataset was originally provided by Olist, the largest department store in Brazilian marketplaces. Olist connects small businesses across Brazil to major sales channels through a single contract. Merchants can sell their products on the Olist Store and ship them directly to customers using Olist's logistics partners. You can learn more about Olist at www.olist.com.

When a customer makes a purchase on the Olist Store, the seller is notified to fulfill the order. Once the product is delivered—or once the estimated delivery date has passed—the customer receives a satisfaction survey by email, where they can rate their purchase experience and leave comments.

Challenge Tasks

1. Data Pipeline

Your task is to design a data architecture for an analytics platform using Google Cloud Platform. Consider the following:

- Data Sources:
 - RDBMS
 - Files (CSV, Sheet, Parquet, Avro)
 - Partner APIs
 - Streaming systems (Kafka, Pub/Sub)
- Data Warehouse:
 - BigQuery
- Presentation:
 - Reports
 - Dashboards
 - Alerts
 - Analytical results

Instructions:

Please design the data platform architecture covering ingestion, storage, transformation, and presentation. Mention the tools you would use at each stage and explain why you chose them.

2. Query and Data Modeling

- Cleanse the provided dataset.
- Create a data warehouse using the dataset.
- Provide the ERD for your data warehouse.
- Provide SQL scripts used for creating the data warehouse and performing data transformations.
- Build a simple API to retrieve data from the data warehouse.

You are free to use any technology and design approach for the data warehouse. Be creative! your imagination is the limit.

If you have any questions about the data or the test, you can always reach out to us at:

annissa.putri@stockbit.com or data-engineer-group@stockbit.com

Please submit your answers in a ZIP file named:

Your Name – Data Engineer Test

Upload your test files here: <https://bit.ly/testsubmissionDE>

Good luck!